Elective Surgery Scheduling

An interval-based constraint program for weekly OR planning

Hector Bonilla

Agenda

1. Problem & Motivation
2. Scope & Assumptions
3. The CP-SAT Model, in Depth
4. Parameter Calibration
5. Why CP-SAT, Not a Bigger MIP
6. Architecture & Reusability
7. Open Questions & Next Steps

CP-SAT's own math is fully worked out in the main flow. The comparison MIP stays a backup appendix – it's evidence for the CP-SAT choice, not a second model to present.

The problem

One hospital, one week, an elective waiting list bigger than the week's capacity.

- Decide, for each waiting case: which room, what time, which day, which surgeon – or leave it for a later week.
- Has to be **feasible**: rooms, surgeons, and a handful of shared resources are all finite.
- Has to be **fair**: a genuinely urgent patient shouldn't sit behind a routine one just because the routine case reached the front of a FIFO queue first.

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Brief's own framing: room hours, surgeon availability, shared equipment, turnover between cases, downstream bed pressure – explicitly invites simplification, not a full hospital simulation.

Why it matters – this isn't a toy problem

- **OR time is expensive and perishable.** $\sim$ \$30–40 per minute once staffing/equipment/overhead are amortized, one widely cited U.S. estimate (Macario, 2010) – and an idle OR-minute today can't be banked for tomorrow.
- **This is a recognized sub-problem.** Cardoen, Demeulemeester & Beliën (2010) frame "advance scheduling" – deciding which case runs on which day – as distinct from same-day sequencing or disruption handling. That's the scope of this model.
- **Elective and emergency demand compete for the same rooms** (Van Riet & Demeulemeester, 2015) – the reason an emergency-add-on priority tier exists in the model at all.
- **The waiting list itself is the audited number** – not just a general claim. Next slide: the specific system this model's priority mechanism is grounded in.

The Portuguese case: SIGIC

Sistema Integrado de Gestão de Inscritos para Cirurgia – Portugal's national system for managing the elective surgical waiting list, established by *Portaria n.º 45/2008*.

- Every elective case is assigned one of **four clinical priority tiers**, each with its own maximum acceptable wait: **270 / 60 / 15 / 3 days**.
- This is exactly the same tier structure used by the UK NHS's Referral-to-Treatment targets and several Canadian provincial wait-time benchmarks: tiered deadlines, audited breach rates – not a single undifferentiated FIFO queue.

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Why this matters here: these four numbers are not illustrative defaults invented for this demo – they are *this model's* maximum-wait parameters, wl_p^{max}, taken directly from a real, audited policy.

The Portuguese case: what the 2016 audit found

Marques & Captivo (2015) studied the Centro Hospitalar Lisboa Norte (CHLN) waiting list against the SIGIC targets above:

- ∼**7,400** patients on the list at audit time.
- **16%** had *already* breached their tier's deadline.
- Among breached cases, the average overrun was **147 days** hospital-wide – and ∼**261 days** specifically in Neurosurgery, the worst-performing service.

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The point for this model

Breach penalties in the objective (w_c, Term 3) aren't a cosmetic addition – they encode *exactly* the number a real system using this policy is actually graded on. A model that only maximizes throughput could report a great utilization number while quietly reproducing this audit's failure mode.

What's kept, what's left out

Kept: case→room→time assignment, surgeon availability & workload limits, room/service rosters, one shared piece of equipment, a downstream recovery/ICU-bed constraint.

Left out, deliberately:

- Stochastic durations
- Same-day disruption
- Nurses/anaesthetists as separate resources
- Sequence-dependent cleaning
- Multi-week planning
- An explicit emergency capacity reserve

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Key assumptions (1/2)

1. **Deterministic durations.** One estimated duration per case. Real durations are stochastic and biased *long* – the standard tractable starting point (Denton et al., 2010); stochastic extension named, not attempted.
2. **Turnover depends on the case's own duration**, bucketed (15/25/40 min), not on which two cases are adjacent. The sequence-dependent piece is real but secondary here (CP Optimizer appendix shows how it would work).
3. **Surgeons are the binding staffing resource.** Nurses/ anaesthetists assumed bundled with the room's roster – true whenever the surgeon's calendar, not support staff's, is the actual bottleneck.
4. **One occurrence per patient per week.** Conservative default; combining same-day multi-procedure cases is service-specific clinical judgment, not assumed here.

Key assumptions (2/2)

5. **Bed capacity is constant across the week**, with an *explicit* overflow penalty for a stay crossing the horizon boundary – not a silent approximation either way.
6. **One ad hoc institutional rule (pediatric block)** included as a worked example: costs nothing structurally, just one more eligibility predicate.
7. **Single week, single hospital, no same-day disruption, no explicit emergency reserve.** A genuinely different problem (reactive rescheduling) from advance scheduling.

Full reasoning, trade-offs, and what relaxing each one would require: FORMULATION.md §2.

Sets and indices

Symbol	Meaning
$c \in C$	Surgical cases (one per patient-procedure pair)
$d \in D$	Planning days, $D = \{1, \ldots, 5\}$
$r \in R$	Operating rooms
$h \in H$	Surgeons
$e \in E$	Shared equipment types
$D_c \subseteq D$	Days c may run on ($\{1\}$ if must-run-today)

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Decision variables – presence and disposition

For every (c, d, r) surviving eligibility (room-service roster, ambulatory-only, pediatric block, surgeon availability):

$$\mathrm{pr}_{cdr} \in \{0, 1\} \qquad \mathrm{start}_{cdr}, \mathrm{end}_{cdr} \in [0, k_{dr}]$$

For every case that isn't priority-4, an unscheduled flag:

$$u_c \in \{0, 1\}, \qquad \sum_{d,r} \mathrm{pr}_{cdr} + u_c = 1$$

Candidates are only generated for (c, d, r) triples that pass eligibility – the model's main variable-count reduction, done once up front, never re-checked by a constraint later.

Decision variables – why *two* intervals

$\mathrm{iv}_{cdr} = \texttt{NewOptionalIntervalVar}(\mathrm{start}_{cdr},\ t_c^{tot},\ \mathrm{end}_{cdr},\ \mathrm{pr}_{cdr})$ ROOM, size $t_c^{op} + t_c^{clean}$

$\mathrm{sgiv}_{cdr} = \texttt{NewOptionalIntervalVar}(\mathrm{start}_{cdr},\ t_c^{op},\ \mathrm{sgend}_{cdr},\ \mathrm{pr}_{cdr})$ SURGEON, size t_c^{op}

Both share `start`; both are *present* exactly when $\mathrm{pr}_{cdr} = 1$.

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Worked example

Surgeon H operates Case X in Room 1: $t_c^{op} = 70$, $t_c^{clean} = 20$, starting at $t = 0$. The room stays blocked until $t = 90$. The surgeon's own involvement ends at $t = 70$ – they can scrub into Room 2 for Case Y while nursing staff finish cleaning Room 1. One interval sized t_c^{tot} for C7 and a *separate* one sized t_c^{op} for C8 is what lets the model see that; collapsing both onto the room's longer interval would wrongly forbid it.

Objective function

$$\min \underbrace{\sum_{c:dd_c \geq 0} \sum_{d \in D_c, r} [dd_c + d]\, \mathrm{pr}_{cdr}}_{\text{on-time, prefer earlier}} + \underbrace{\sum_{c:dd_c < 0} \sum_{d \in D_c, r} [dd_c + \alpha d]\, \mathrm{pr}_{cdr}}_{\text{overdue, urgency-weighted}}$$

$$+ \underbrace{\sum_{c:p_c \neq 4} w_c\, u_c}_{\text{non-scheduling penalty}} + \underbrace{\pi^{ovf} \sum_{c:\rho(c) \neq \text{none}} \text{overflow}_c}_{\text{bed-overflow penalty}}$$

$$w_c = \mu_{p_c} \cdot \text{PenaltyCurve}(dd_c) + 1.2 \cdot \max_{c'} dd_{c'}$$

`PenaltyCurve` is flat while a case has slack, then escalates sharply past the deadline – the same shape the SIGIC-style systems above use to make breaches expensive, not merely "less preferred". Term 3 is built to dominate Terms 1–2 (proof: §5 of this deck) – the model only drops a case when there's genuinely no room.

Constraints C1–C3 – case disposition

C1 – one occurrence per patient per week:

$$\sum_{c:\text{patient}(c)=n} \sum_{d,r} \text{pr}_{cdr} \leq 1 \qquad \forall n$$

C2 – priority-4 locked to day 1:

$$\sum_{r} \text{pr}_{c,1,r} = 1 \quad \text{(every other slot for } c \text{ forced to 0)}$$

C3 – schedule-or-penalize: already written into the u_c definition – $\sum_{d,r} \text{pr}_{cdr} + u_c = 1$.

C4–C6 (room-service roster, ambulatory-only, the pediatric block) are resolved during candidate generation, before any of the above – they never appear as a constraint row at all.

Constraints C7–C9 – room and surgeon

C7 – room, exact non-overlap:

$$\texttt{AddNoOverlap}\big(\{\mathrm{iv}_{cdr} : d, r \text{ fixed}\}\big) \qquad \forall d, r$$

C8 – surgeon, exact non-overlap on their own window, plus a daily cap:

$$\texttt{AddNoOverlap}\big(\{\mathrm{sgiv}_{cdr}\}\big), \qquad \sum_{c:\mathrm{surgeon}(c)=h} \sum_{r} t_c^{op}\, \mathrm{pr}_{cdr} \leq k_{hd}$$

C9 – surgeon weekly limit:

$$\sum_{c:\mathrm{surgeon}(c)=h} \sum_{d,r} t_c^{op}\, \mathrm{pr}_{cdr} \leq k_h \qquad \forall h$$

NoOverlap alone bounds concurrency, not total hours – C8/C9's minute caps are kept alongside it for exactly that reason.

Constraints C10–C11 – equipment and beds

C10 – shared equipment, exact concurrency:

$$\texttt{AddCumulative}\big(\{\mathrm{iv}_{cdr} : u_{ce} = 1,\ d \text{ fixed}\},\ 1,\ \kappa_{ed}\big) \qquad \forall e, d$$

C11 – recovery/ICU beds, day-granularity, with overflow:

$$\mathrm{bed}_c = \texttt{NewOptionalIntervalVar}(\mathrm{dayof}_c,\ \mathrm{los}_c,\ \mathrm{dayof}_c + \mathrm{los}_c,\ 1 - u_c)$$

$$\texttt{AddCumulative}\big(\{\mathrm{bed}_c : \rho(c) = \rho\},\ 1,\ \beta_\rho\big) \qquad \forall \rho$$

$$\mathrm{overflow}_c = \max\big(0,\ (\mathrm{dayof}_c + \mathrm{los}_c) - n_{days}\big)$$

A multi-day stay needs a real "day of surgery" value to start counting from – the concrete reason this model is interval-based at all, not just a stylistic choice.

Agenda

How every default was chosen – four honest categories

1. **Literature-grounded** – taken from a real, published policy or study (e.g. SIGIC's max-wait days).
2. **Derivable rule** – computed from other parameters by an explicit, checkable formula (e.g. the priority multipliers).
3. **Demo-design choice** – sized specifically to make the demo data exercise a constraint, not measured from any one hospital.
4. **Placeholder** – stands in for data a real deployment must supply (e.g. ICU admission probability in the synthetic generator).

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Every one of these lives on `PlanningInstance` (`src/model/types.py`), never hardcoded in solver logic – a hospital overrides its own policy without touching the model.

Turnover and time budgets

Room turnover – 15/25/40 min by duration bucket. Real OR turnover is reported anywhere from 15 to 60 minutes depending on procedure complexity and infection-control needs; bucketing by the case's own duration is a simple proxy for that spread (category 3/4 – a real deployment should use measured hospital data instead).

Surgeon limits – 240 min/day in the demo data is one standard half-day theatre session, a common unit in block-scheduling literature (Cardoen et al., 2010); 960 min/week leaves room for clinics, ward rounds, and on-call duties across the rest of a typical job plan (category 1/2: structurally grounded, not a single cited number).

Max waits and priority multipliers

Max waits (270/60/15/3 days) – Portugal's SIGIC policy, directly (category 1: literature-grounded – this is the system presented two slides ago).

Priority multipliers ($\mu_p = 1$, 4.5, 18, 90) – *derived*, not invented (category 2):

$$\mu_p = \frac{\mathrm{wl}_1^{max}}{\mathrm{wl}_p^{max}} \;\Rightarrow\; \mu_2 = \frac{270}{60} = 4.5, \quad \mu_3 = \frac{270}{15} = 18, \quad \mu_4 = \frac{270}{3} = 90$$

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A tier allowed only a twelfth of priority-1's wait (15 vs. 270 days) is, by the *same* policy's own design, roughly twelve times more wait-sensitive – weighting its breach days twelve times as heavily follows from the policy that set the wait targets, not a second, independent judgment call.

The urgency multiplier α and the displacement margin

$\alpha = 2.0$ appears only in Term 2's day coefficient – it can only change *which day* an overdue case lands on, never *how many* get scheduled (capacity and Term 3 govern that, and neither involves α).

Term 3 (w_c) must dominate every Term-1/2 coefficient, or the model could prefer dropping a schedulable case to dodge a tardiness charge:

$$\text{margin}_{\text{min}} = 1 + \frac{\alpha \cdot n_{days}}{\max_c dd_c} \approx 1.037 \quad (\alpha = 2,\ n_{days} = 5,\ \max_c dd_c \approx 270)$$

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The model uses **1.2** – clears the minimum with real margin, since $\max_c dd_c$ shrinks on instances with only short-horizon, high-priority cases, and a fixed margin has to stay safe across that range (category 2: derived, with an explicit safety buffer on top).

Equipment and bed capacity

The demo's shared C-arm has capacity **1**, deliberately tight enough to force real contention among the four cases that need it – the point is to make the CP-vs-MIP gap (next section) actually visible, not to model one hospital's real inventory (category 3).

ICU bed capacity (2/day demo, 6/day medium) follows the same logic: tight enough that long-stay cases create real pressure without making the instance infeasible. The overflow penalty $\pi^{ovf} = 50/\text{day}$ sits between two things it has to balance – big enough to matter against a typical day-coefficient swing, small enough that it never beats simply not scheduling the case (which costs w_c, in the hundreds to low thousands).

The 12% ICU-admission rate in the synthetic generator is category 4 – a placeholder for real admission data, never seen by the optimizer as a parameter at all.

The core argument

"Total minutes used today $\leq$ capacity" is not the same statement as "these cases don't collide"

For one room, a set of non-overlapping durations can always be packed sequentially – the two statements coincide. They stop coinciding the moment a resource is *shared across rooms* (this project's C-arm).

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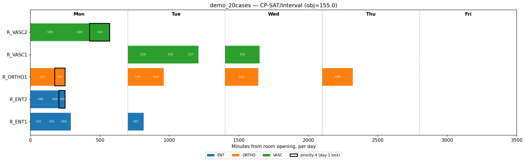
- `NoOverlap` / `Cumulative` are global constraints with their own polynomial-time propagation (Vilím, 2004; Schutt et al., 2009) – they prune from the problem’s time structure directly.
- The MIP alternative (pairwise big-M disjunctions) is quadratic in conflicting pairs, and the big-M constants weaken the LP relaxation as that count grows (Baptiste, Le Pape & Nuijten, 2001).
- CP-SAT’s parallel portfolio search is the actual state of the art here – no hand-rolled branching strategy on top of it.

▶ Full comparison MIP formulation (backup)

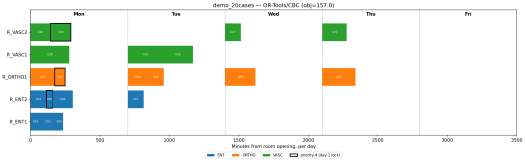
Proof, not assertion – the demo instance

One shared C-arm, capacity 1, needed by 4 cases.

Solver	Status	Objective	Scheduled
CP-SAT (primary)	Optimal	**155.0**	20/20
OR-Tools/CBC (comparison MIP)	Optimal	157.0	20/20



CP-SAT: two C-arm cases share Tuesday

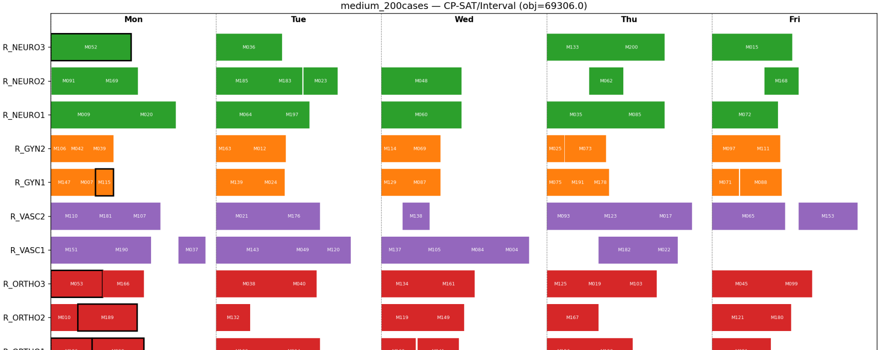


Comparison MIP: one C-arm case per day, forced

CP-SAT checks literal time overlap and legally puts two C-arm cases on the *same* day; the MIP's day-count cap forbids that outright – a smaller feasible region, not a worse search.

Proof at scale – 200 cases, 12 rooms, 17 surgeons

Solver	Status	Objective	Own gap	vs. true MIP opt.	Sched.
Gurobi (true MIP optimum)	Optimal	74,074.0	0.00%	—	130/200
OR-Tools/CBC (2 min)	Feasible	74,116.0	0.56%	+0.06%	130/200
CP-SAT (2 min)	Feasible	**69,956.0**	6.62%	−5.56%	**131/200**



How it's built

- **CP-SAT** (`cp_sat_interval_solver.py`) – the model, default and only backend needed to see it run.
- **Comparison MIP** (`milp_baseline_solver.py`) – CBC / SCIP / Gurobi / CPLEX, kept *only* as the empirical check on the CP-over-MIP argument – full math in the backup appendix, not presented as a second model.
- **CP Optimizer** (`cp_optimizer_solver.py`, optional, license-gated) – a second CP engine that can model sequence-dependent turnover; falls back to CP-SAT automatically.
- `tests/test_model.py` – the acceptance contract every solver's output is checked against (hard constraints, cross-solver agreement on the demo instance).

A reusable library of models

Four layers, solver-agnostic except the bottom one:

1. **Core data types** – plain dataclasses, no solver imports.
2. **Constraint patterns** – capacity sums, NoOverlap-based non-overlap, tiered-priority tardiness, eligibility pre-filters – reusable across nurse rostering, bed allocation, this project.
3. **Problem templates** – compose those patterns into a specific formulation. This project is one template.
4. **Solver-adapter layer** – one file per backend family (MILP, CP, local search), so a new problem picks a backend without rewriting how constraints are expressed.

The CP-vs-MIP comparison this project runs end to end is itself the template for layer 4: argue the backend choice from the problem's structure, then check it empirically, before committing.

Passing the torch

- FORMULATION.md + FORMULATION_CP.md – the math, one source of truth.
- `src/model/types.py` – the dataclasses *are* the data dictionary; every symbol maps to a field.
- The solver code – every constraint carries the same C-number as the math, read side by side.
- `tests/test_model.py` – the acceptance bar; a new test alongside any new constraint, not after it.
- A short glossary – most confusion on a project like this is vocabulary ("room roster", "ambulatory"), not the math.

Where I'd take this next

Extension	Approach
Stochastic durations	Two-stage stochastic program
Same-day rescheduling	LNS seeded from the current plan
Explicit emergency reserve	Carve out a per-day capacity block
Nurse/anaesthetist rostering	Extend H, reuse the surgeon pattern
Multi-week rolling horizon	Solve weekly, carry forward at bumped priority
Day-varying bed capacity	Per-day-segmented cumulative resource
Per-specialty fairness	Secondary objective bounding overdue share

Questions

Thank you.

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Backup G – the comparison MIP, in full

Same sets and shared parameters as the CP-SAT model. Variables:

$$x_{cdr} \in \{0,1\} \ \forall c, d \in D_c, r \qquad z_c \in [0,1] \ \forall c : p_c \neq 4$$

Objective – identical shape to CP-SAT's Terms 1–3:

$$\min \sum_{c:dd_c \geq 0} \sum_{d,r} [dd_c + d]\, x_{cdr} + \sum_{c:dd_c < 0} \sum_{d,r} [dd_c + \alpha d]\, x_{cdr} + \sum_{c:p_c \neq 4} w_c\, z_c$$

Same C1–C6, C9 as CP-SAT (room-service roster etc. pre-filtered the same way). Diverges at C7 and C10.

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Backup G – where the MIP diverges, and why there's no C11

C7 – room capacity as a sum, not exact non-overlap:

$$\sum_{c} t_c^{tot}\, x_{cdr} \leq k_{dr} \qquad \forall d, r$$

For a single room this is equivalent to exact non-overlap – any non-colliding set of durations can always be packed sequentially.

C10 – shared equipment, a day-level headcount, not exact overlap:

$$\sum_{c:u_{ce}=1} \sum_{r} x_{cdr} \leq \kappa_{ed} \qquad \forall e, d$$

This counts *how many* equipment-e cases land on a day, not whether their clock times overlap – the single largest source of the objective gap measured earlier.

No C11. A day-bucket model has no value that means "day of surgery" for a multi-day bed stay to start from. This, more than the equipment gap, is the structural reason this project ended up CP-based.

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Backup H – CP Optimizer (optional second CP engine)

Kept for one reason: sequence-dependent turnover, which CP-SAT can't express without restructuring its intervals.

	Same service, back-to-back	Different service, back-to-back
CP-SAT (duration-bucketed)	charged on the case alone	same
CP Optimizer (transition matrix)	15 min	35 min

Demo instance: identical objective to CP-SAT (155.0, 20/20). At the 200-case scale its own gap closes far more slowly – a search-tuning gap (no warm start applied), not a modeling one. Stays an appendix backend, not the model.

Agenda

Backup I – references

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